Hasan Shaikh

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Work Experience

Quantitative Imaging Research and Artificial Intelligence Lab (QIRAIL) Tamil Nadu, India

Project Assistant, Christian Medical College Vellore, Dept. of Radiation Oncology *Aug. 2024 – Present*

[P1] Large-Scale Prospective Imaging Study – DBT/Wellcome Trust India Alliance

- Contributed to the end-to-end radiomics pipeline for **1,900+ head and neck cancer patients** in a DBT/Wellcome Trust-funded prospective study (2020-2025) — DICOM retrieval, automated tumor segmentation, and PyRadiomics feature extraction.
- Deployed AWS S3 backup pipelines with error recovery, and helped coordinate annotation workflows across radiation oncology sites.
- **Tech Stack:** Python, Orthanc PACS, DICOM, PyRadiomics, Citric, Bash

[P2] Radiomics-Based Risk Stratification in Head and Neck Cancer

- Developed a 10-feature prognostic signature (2 clinical + 8 radiomics) from 1000+ candidate features, achieving **AUC 0.82 (95% CI: 0.62–0.95)** for locoregional recurrence prediction in 163 patients.
- Benchmarked 5 metaheuristic feature selection algorithms — Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Grey Wolf Optimization (GWO), Whale Optimization Algorithm (WOA), and Simulated Annealing (SA) — across multiple classifiers to find which works best for high-dimensional radiomics data.
- **Tech Stack:** PyRadiomics, scikit-learn, NumPy, Pandas, SciPy, Metaheuristic Algorithms

[P3] Automated Segmentation of Head and Neck Cancer Using 3D CNNs – Collaboration with NIT Surathkal

- Curated a multi-institutional dataset (167 cases: 137 MAASTRO + 30 CMC) and found a **7% performance drop** on Indian patients relative to European data, surfacing a critical generalization gap.
- Trained **3D nnU-Net** for CT-based segmentation, achieving **Dice: 0.62 (MAASTRO), 0.63 (CMC), 0.65 (combined)**, demonstrating measurable gains from cross-institutional training.
- **Tech Stack:** nnU-Net, DICOM, NIfTI, ITK-SNAP, 3D Slicer

[P4] CHAVI: CompreHensive Digital ArchiVe of Cancer Imaging – India's First National Oncology Biobank

- Contributed **304+ de-identified cases** with validated clinical and imaging metadata to India's first national oncology imaging biobank (collaboration with Tata Memorial Centre and IIT Kharagpur), using DICOM and XNAT pipelines.
- **Tech Stack:** XNAT, Orthanc, Data Anonymization Tools

[P5] Reproducibility Study: CNN-Based Head and Neck Cancer Prognosis

- Audited a published CNN model for HNC prognosis across three outcomes (distant metastasis, locoregional failure, overall survival); identified incomplete datasets, undocumented preprocessing, and broken validation protocols.
- Corrected all identified errors and successfully reproduced the results; the original authors formally acknowledged the issues, contributing to improved reproducibility standards in published medical AI.
- **Tech Stack:** NumPy, Pandas, GitHub, Docker

STARlab Capital Lucknow, India
Research Analyst *Dec. 2023 – June 2024*

- Designed and backtested volatility-based options strategies (Nebula, ARUT, A2) using **OptionNet Explorer, Mesosim, OptiTrade, OptiBot** tools.
- Led contributions to OptiTrade's open-source repository, improving internal backtesting and trade log systems.

Publications & Research

Conference Publications

[1] Balu Krishna S, Amal Joseph Varghese, **Hasan Shaikh** et al., *"Development and validation of a prospective radiomics-clinical signature for locoregional recurrence in patients with locally advanced head and neck cancer"*, *European Society for Radiotherapy and Oncology (ESTRO)*, Stockholm, Sweden, 2026. [**Accepted as Poster Highlight**]

[2] **Hasan Shaikh**, Balu Krishna S, Amal Joseph Varghese, et al., *"Metaheuristic-Driven Machine Learning Pipelines for Radiomics-Based Prediction of Locoregional Recurrence in Head and Neck Cancer"*, In: *Proceedings of the International Conference on Artificial Intelligence for Healthcare (AIHC), Lecture Notes in Electrical Engineering, Springer Nature*, 2025. [**Accepted**]

Preprints

[1] Piyus Prabhanjans, Asjad Nabeel P, Aparna V K, Hannah Mary Thomas T, Balu Krishna Sasidharan, **Hasan Shaikh**, Amal Joseph Varghese, Rajendra Benny Kuchipudi, Simon Pavamani, Jeny Rajan, *"Automated Segmentation of Head and Neck Cancer from CT Images Using 3D Convolutional Neural Networks"*, *medRxiv* (preprint), 2026. https://doi.org/10.64898/2026.03.12.26347996

Book Chapters

[1] **Hasan Shaikh** and Rashid Ali, *"Cancer Survival Prediction Using Artificial Intelligence: Current Status and Future Prospects"*, *Data Science in the Medical Field, Academic Press, Elsevier*, 2024. https://doi.org/10.1016/B978-0-443-24028-7.00016-7

Poster Presentations

[1] **Hasan Shaikh**, Amal Joseph Varghese, Balu Krishna S, et al., *"Before We Treat, Can We Tell? A Locoregional Recurrence Signature in Head & Neck"*, *15th Research Day*, Christian Medical College, Vellore, Tamil Nadu, India, 2025.

[2] **Hasan Shaikh**, Amal Joseph Varghese, Hannah Mary Thomas T et al., *"Can CT Radiomics Predict Recurrence in Head and Neck Cancer? Early Results from a Prospective Imaging Trial"*, *14th Research Day*, Christian Medical College, Vellore, Tamil Nadu, India, 2024.

Master's Thesis

Hasan Shaikh, *"Multimodal Data Analytics for Predicting the Survival of Cancer Patients"*, Aligarh Muslim University, 2023. Advisor: Prof. Rashid Ali.

Implemented **MGAttCNNMD** — a multimodal gated attention CNN integrating clinical records, gene expression, and copy number alterations — achieving **91.2% survival prediction accuracy** on the METABRIC breast cancer dataset.

Technical Skills

Machine Learning & DL	: TensorFlow, Keras, scikit-learn, PyRadiomics, nnU-Net
Medical Imaging	: 3D Slicer, ITK-SNAP, Orthanc PACS, XNAT, DICOM, NIfTI
Programming Language	: Python (NumPy, Pandas, SciPy), Bash
Data Pipeline & Deployment	: Docker, GitHub
Specialized Methods	: Radiomics, Metaheuristic Techniques, Graph Neural Networks

Education

Master of Technology in Computer Engineering — Nov. 2021 – Nov. 2023
Aligarh Muslim University (AMU), CGPA: 8.80 / 10.00 — *Aligarh, India*

Bachelor of Technology in Computer Science and Engineering — Aug. 2017 – Jul. 2021
Dr. A.P.J. Abdul Kalam Technical University (AKTU), CGPA: 8.04 / 10.00 — *Lucknow, India*

Academic Achievements & Honors

Third Prize, Oral Presentation at the 2nd National Symposium on Health Data and AI (March 2025) — Presented *"Automated Segmentation of Head and Neck Cancer from CT Images Using 3D nnU-Net"*, awarded by the BioMedical Informatics Unit, CMC Vellore.

Honors Degree — Ranked in the top 1% out of 128 students in the undergraduate engineering program.

Completed NPTEL courses conducted by IITs (Elite + Silver Certified):

- Data Analytics with Python (80%) (Link)
- Essential Mathematics for Machine Learning
- Deep Learning
- Python for Data Science (78%) (Link)
- Machine Learning
- Demystifying the Brain (Link)

Relevant Courses

Relevant courses completed during my Bachelor's and Master's programs include:

- Artificial Intelligence
- Machine Learning
- Web Mining & Searching
- Big Data Analytics
- Image Processing & Computer Vision
- Application of Soft Computing
- Data Structures
- Applied Linear Algebra
- Design & Analysis of Algorithms
- Database Management Systems
- Data Warehousing & Data Mining
- Distributed Systems

Attended Conferences & Workshops

1. Presented paper at the **International Conference on Artificial Intelligence for Healthcare (AIHC)**, organized by National Institute of Technology (NIT) Calicut, India (December 10 - 12, 2025) on Metaheuristic-driven radiomics-based locoregional recurrence prediction in head and neck cancer. [*Lecture Notes in Electrical Engineering, Springer*] (Link)
2. Organized the **1st & 2nd Workshop on Radiomics and Auto Segmentation**, conducted by the Department of Radiation Oncology & Quantitative Imaging Research and Artificial Intelligence Lab, Christian Medical College (CMC) Vellore, Tamil Nadu, India (November 14–15, 2025 & March 27-28, 2026). (Link)
3. Organized the **Head and Neck Cancer Survivorship Program**, conducted by the Department of Radiation Oncology, Christian Medical College (CMC) Vellore, Tamil Nadu, India (September 12–13, 2025). Themes: "Together We Overcome, Together We Heal" and "De-Escalation Strategies in Head and Neck Cancer."
4. Participated and served as part of the Organizing Team for the **2nd Annual Winter Symposium on Health Data and AI**, conducted by Biomedical Informatics Unit, Christian Medical College (CMC) Vellore, Tamil Nadu, India (March 13–15, 2025).
5. Attended the Continuing Medical Education (CME) program on **Revolution and Precision in Radiation Oncology**, Ida B. Scudder Cancer Center, Christian Medical College (CMC) Vellore, Tamil Nadu, India (March 1, 2025).
6. Participated in the **14th & 15th Annual Research Day**, organized by the Office of Research, Christian Medical College (CMC) Vellore, Tamil Nadu, India (October 24–25, 2024 & 2025).
7. **AI & ML for Engineering & Social Sciences Research**, 2023 IEEE Computational Intelligence Society (CIS) Summer School, organized by Malaviya National Institute of Technology Jaipur (MNIT) Jaipur, India, 4 – 8 Sep. 2023. (Link)
8. **7th Summer School on AI with Focus on Computer Vision & ML**, coordinated by International Institute of Information Technology (IIIT) Hyderabad, India, 1 – 31 Aug. 2023. (Link)

9. **Emerging Research Trends in Computational Intelligence Techniques to Address Challenges in Biomedical Data and Imaging**, 2022 IEEE CIS Summer School, organized by National Institute of Technology (NIT) Arunachal Pradesh, India, 7 – 11 Nov. 2022. (Link)

Referees

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